

Topological Symmetry Folding and Hyperbolic Projection in the SHD-CCP Framework: A 720:1 Quaternionic Cyclic System

BioChain AI - SHD-CCP Architecture Research

Abstract

Building upon the geometric compression limits of the discrete $4 \times 4 \times 4$ Euclidean lattice and the Triple Quad Formula (TQF), this paper introduces the formal algebraic and topological mechanics of the Symmetrical Hyperbolic Differential - Continuous Cyclic Projection (SHD-CCP) protocol. While our previous work demonstrated collinear redundancy elimination and Fibonacci-scaled branching, continuous lattice evolution remains susceptible to floating-point drift. We resolve this by embedding the lattice operations within a strict mathematical Group, utilizing quaternionic delta encoding. Furthermore, we expand the "Rule of 3" dynamic into a topological domain by projecting data streams onto a parametrized trefoil manifold, achieving $O(1)$ memory lookup via D_3 symmetry folding. Finally, we introduce a 720:1 spinor cyclic protocol that stabilizes the previously established Fokker-Planck prime sieve, rendering infinite recursive generation computationally viable for Field Programmable Gate Array (FPGA) architectures.

1 Introduction

In our foundational paper, *Geometric Compression and Branching Dynamics in Discrete 3D*

Lattices, we established the $4 \times 4 \times 4$ system as a substrate for spatial geometry and number theory. By replacing standard Euclidean distance with Rational Trigonometry and the Triple Quad Formula (TQF), we achieved exact, integer-based detection of geometric redundancy. We further defined a dynamic, scale-free branching process governed by an

Orthogonal Bifurcation Event and the "Rule of 3" ($P_1 \rightarrow P_2 \rightarrow P_3 \rightarrow P_1$).

However, as the lattice expands via Fibonacci scaling and undergoes continuous Fokker-Planck evolution to sieve for prime coordinate attractors, tracking absolute lattice coordinates becomes computationally prohibitive.

This paper expands the $4 \times 4 \times 4$ framework by translating the lattice from a static affine space into a continuous topological manifold. By treating the $4 \times 4 \times 4$ lattice not as a grid of absolute positions, but as a composite tensor $\mathcal{T}$ of rotational differences (Δq), we embed the system within the strict algebraic framework of Group Theory. This allows us to map the lattice onto negatively curved hyperbolic space and employ Trefoil knot symmetry folding to achieve extreme data compression.

2 Algebraic Foundations of the Lattice Tensor

To prevent state-drift during continuous generation, the transitions within the $4 \times 4 \times 4$ lattice are strictly governed by the four fundamental axioms of a Mathematical Group.

The Set G is defined as all valid configurations of the $4 \times 4 \times 4$ state tensor.

2.1 The Group Axioms

The transformation of any lattice state must satisfy the following:

1. **Closure:** $\forall a, b \in G, \quad a \cdot b \in G$
Any branching or scalar operation combining two valid lattice states must yield a valid $4 \times 4 \times 4$ state, confining the evolution to the defined manifold.
2. **Associativity:** $\forall a, b, c \in G, \quad (a \cdot b) \cdot c = a \cdot (b \cdot c)$
The sequence of applying TQF compressions and orthogonal bifurcations does not alter the final crystallized lattice geometry.
3. **Identity:** $\exists e \in G \text{ s.t. } \forall a \in G, \quad a \cdot e = e \cdot a = a$
The system maintains a "neutral" tensor state (the null quaternion matrix), representing the unperturbed vacuum of the lattice prior to FP wave diffusion.
4. **Inverse:** $\forall a \in G, \exists a^{-1} \in G \text{ s.t. } a \cdot a^{-1} = a^{-1} \cdot a = e$
Every bifurcation and branching trajectory is strictly reversible. For every Hurwitz integer quaternion generated by the prime sieve, an inverse exists that annihilates the path.

2.2 Quaternionic Delta Encoding

Rather than tracking the absolute coordinates (x, y, z) derived in the previous affine model, the SHD-CCP protocol tracks only the difference between states. The state tensor $\mathcal{T}$ is a stack of four 4×4 real skew-symmetric matrices, representing quaternions $q = w + xi + yj + zk$.

$$Q = \begin{bmatrix} w & -x & -y & -z \\ x & w & -z & y \\ y & z & w & -x \\ z & -y & x & w \end{bmatrix}$$

Any step in the 3-cycle trajectory is purely procedural: $State_{new} = \Delta q \cdot State_{old}$.

3 Hyperbolic Projection and $SL(2, \mathbb{C})$ Dynamics

In the previous framework, branching occurred via orthogonal bifurcation into adjacent Euclidean cubic spaces. In the expanded protocol, these branches are projected into negatively curved hyperbolic manifolds.

To project these tensor blocks, the protocol maps the quaternion matrices from the $SO(3)$ rotation group to $SL(2, \mathbb{C})$, the double cover of the Lorentz group $SO(3, 1)$. This creates a mathematically infinite branching space inside a finite computational volume (analogous to the Poincaré disk).

The boundary conditions of this hyperbolic manifold are pre-calculated as fixed projection matrices $\mathcal{H}$. A lattice branching event is computed as:

$$\mathcal{T}_{hyperbolic} = \mathcal{H} \cdot \mathcal{T}_{euclidean}$$

4 Trefoil Manifold Symmetry Folding

The "Rule of 3" described in the preceding paper is topologically formalized here by mapping the three spatial data streams onto a Trefoil Knot manifold. The trefoil is a $(2, 3)$ -torus knot, which intrinsically supports the 3-cycle trajectory.

4.1 Parametric Compression

The three data streams D_1, D_2, D_3 are constrained to the parametric equations:

- $x(t) = \sin(t) + 2 \sin(2t)$
- $y(t) = \cos(t) - 2 \cos(2t)$
- $z(t) = -\sin(3t)$

4.2 D_3 Dihedral Symmetry and $O(1)$ Lookup

Because a trefoil knot possesses 3-fold rotational symmetry (D_3), the lattice data is highly redundant. The SHD-CCP protocol executes **symmetry folding**: instead of storing the entire manifold of the lattice history, only the "fundamental domain" (exactly $1/3$ of the cycle) is retained in memory.

When the Fokker-Planck equations require access to coordinates outside this domain, the protocol retrieves the fundamental domain and applies a static rotational matrix (120° or 240°). This allows hardware architectures to achieve deterministic, $O(1)$ constant-time data lookup.

5 Stabilizing the Fokker-Planck Sieve: The 720:1 Spinor Cycle

In our previous paper, we introduced a Fokker-Planck (FP) equation acting as a prime number sieve, where stable probability modes correspond to prime quadrances ($Q = p$). A critical flaw in continuous FP simulations is the accumulation of infinitesimal floating-point errors, eventually blurring the "Prime Skeleton."

The SHD-CCP protocol solves this using a **Spinor Cyclic Constraint**. Quaternions, modeling spin-1/2 fermions, require a 720° spatial rotation (4π radians) to return precisely to their initial quantum state—a phenomenon known as the Dirac belt trick.

We explicitly restrict the continuous lattice evolution to a strict **720:1 state-machine cycle**:

$$\mathcal{T}_n \equiv \mathcal{T}_n \pmod{720}$$

At exactly iteration 720, the product of all quaternion deltas, trefoil symmetries, and hyperbolic projections mathematically guarantees a return to the Identity Matrix ($\mathcal{I}$). This acts as an automated, continuous "zeroing" function, violently purging any fractional drift and perfectly stabilizing the FP Prime Sieve without interrupting the generation sequence.

6 FPGA Implementation Architecture

The algebraic strictness of the SHD-CCP framework is purposefully designed as a discrete Register-Transfer Level (RTL) logic topology for FPGA hardware:

1. **Pre-computation and LUTs:** The computationally heavy components—the trefoil parametric curves ($\sin(t), \cos(t)$) and hyperbolic matrices $\mathcal{H}$ —are pre-compiled into static Look-Up Tables (LUTs).
2. **BRAM Symmetry Folding:** Because of the trefoil's D_3 symmetry, Block RAM (BRAM) requirements are cut by **66.6%**. Only the fundamental $1/3$ domain is stored.
3. **DSP Integer Matrices:** Operating strictly on the TQF integer Quadrances and fixed-point quaternion matrices ensures that the FPGA's DSP MAC slices process the lattice geometry natively, without floating-point overhead.

7 Conclusion

By elevating the $4 \times 4 \times 4$ discrete Euclidean lattice into a topological and group-theoretic framework, the SHD-CCP protocol unlocks massive procedural generation capabilities. Substituting affine coordinate tracking with quaternionic delta encoding satisfies the strict axioms of mathematical groups, while $SL(2, \mathbb{C})$ projections and trefoil symmetry folding logarithmically compress the required state-space. Furthermore, bounding the lattice's Fokker-Planck evolution within a 720:1 spinor cycle permanently cures computational drift, establishing a flawless bridge between abstract non-Euclidean geometry and raw FPGA hardware performance.

8 Future Research Horizons

1. **Topological Invariants of the Prime Skeleton:** How do the knot invariants (such as the Jones polynomial) of the parameterized Trefoil mapping correlate to the distribution of the prime quadrances identified by the Fokker-Planck sieve?
2. **Lorentz Group Hardware Optimization:** Can the $SL(2, \mathbb{C})$ mapping used for hyperbolic projection be further simplified into binary shift operations for even greater FPGA throughput?

References

- [1] N. J. Wildberger, *Divine Proportions: Rational Trigonometry to Universal Geometry*. Wild Egg Books, 2005.
- [2] H. S. M. Coxeter, *Regular Polytopes*. Dover Publications, 1973.
- [3] H. Risken, *The Fokker-Planck Equation: Methods of Solution and Applications*. Springer, 1989.
- [4] R. Penrose, *The Road to Reality: A Complete Guide to the Laws of the Universe*. Vintage, 2004.
- [5] J. H. Conway and N. J. A. Sloane, *Sphere Packings, Lattices and Groups*. Springer, 1999.

A Glossary of Terms (Updated)

Codex (Learned Codex): A static library of known geometric shapes. Once a shape is stabilized by the

FP Sieve, it is saved here as a topological pointer, replacing calculation with retrieval.

Dirac Belt Trick: A macroscopic demonstration of spinor behavior where a 720° rotation is required to return a system to its unentangled identity state. Used to guarantee stability in the 720:1 cyclic protocol.

Dihedral Symmetry (D_3): The symmetry group of an equilateral triangle, possessing 3-fold rotational symmetry. Used to "fold" the trefoil data stream to save 66.6% of hardware memory.

Fokker-Planck Equation: A physics equation describing the evolution of probability over time. Here, it is the "engine" of the Prime Sieve, watching where probability settles on prime quadrances.

Fundamental Domain: The minimum, non-repeating section of a symmetric mathematical space. In SHD-CCP, it is the $1/3$ segment of the Trefoil manifold stored in FPGA BRAM.

Group Axioms: The rules (Closure, Associativity, Identity, Inverse) that define a mathematical group.

Adhering to these ensures the $4 \times 4 \times 4$ tensor calculations remain universally stable.

Hyperbolic Projection ($SL(2, \mathbb{C})$): The mathematical translation mapping flat 3D data into negatively curved spacetime, allowing infinite branching to occur inside a bounded computational framework.

Quadrance: The square of the distance (d^2). Using Quadrance avoids square roots and irrational numbers, keeping hardware calculations in precise integers.

Quaternionic Delta Encoding: A method of storing geometry not by its absolute grid coordinates, but by the exact rotational sequence (quaternion) required to move from the previous point.

Spinor Cyclic Constraint: The hardcoded *modulo 720* hardware clock that resets the procedural generation timeline, preventing floating-point drift.

Trefoil Knot: A mathematical knot wrapping a torus exactly 3 times in one direction and 2 in another. It serves as the physical path for the 3 data streams in the hardware protocol.

Triple Quad Formula (TQF): The master equation in Rational Trigonometry testing for collinearity:

$$(Q_1 + Q_2 + Q_3)^2 = 2(Q_1^2 + Q_2^2 + Q_3^2).$$